

Masud Ali

 github.com/Masudali23  www.linkedin.com/in/masud-ali-637722298  masud.ali.cd.eee23@itbhu.ac.in

EDUCATION

Indian Institute of Technology (BHU) Varanasi	Aug 2023 - Oct 2027
<i>Bachelor of Technology in Electrical Engineering</i>	<i>CGPA: 9.34</i>
Umakanta Academy	Apr 2021 - Jun 2023
<i>TBSE(XII)</i>	<i>92.2%</i>
BaidyarDighi Higher Secondary School	Apr 2021
<i>TBSE(X)</i>	<i>96.20%</i>

TECHNICAL SKILLS

Areas of Interest: Robotics, Deep Learning, Computer Vision, Control Systems, Data Structures and Algorithms.

Languages: C, C++, Python, Arduino, LaTeX, HTML, CSS, XML, URDF.

Tools/Frameworks: Git/GitHub, ROS, Carla, TensorFlow, PyTorch, Gazebo, OpenCV, CAD(Fusion 360).

EXPERIENCE

Global Remote Internship, Changwon National University (South Korea)	Dec 2024 - Feb 2025
<i>Remote Research Intern under Professor Oh-Seol Kwon</i>	<i>Report</i>
• Goal: Development of an advanced multi-modal architecture for 3D object detection in autonomous systems • Developed a multi-modal 3D object detection architecture (Vision Transformer, EfficientDet, 3D CNN, TransCAR) • Implemented Slave/Master Transformers and Adaptive Controller for cross-attention and weight updates • Evaluated on NuScenes, Lyft, and KITTI datasets • Skills: Deep Learning (Transformers, CNNs), Multi-modal Fusion, Python (PyTorch), Computer Vision • Exposure: Global remote collaboration, advanced neural network architecture, real-world data pipelines	

PROJECTS

Quadrotor Stability and Single Motor Failure Recovery	Oct 2024 - Dec 2024
<i>Inter IIT TechMeet 13.0, IIT Bombay, IdeaForge 2024</i>	<i>Report</i>
• Participated in the Inter IIT TechMeet 13.0 at IIT Bombay, addressing the IdeaForge problem statement • Collaborated to achieve stable hovering and controlled landing of a quadrotor with a single motor failure • Implemented sensor fusion using the Extended Kalman Filter (EKF) for accurate state estimation and control • Developed control mechanisms to maintain quadrotor stability during hover and landing phases • Exposure: PX4 Autopilot, EKF, Sensor Fusion, Stability Control, UAV Dynamics, Mavros	
Agricultural Theme Manual and Autonomous Bot	Jan 2024 - July 2024
<i>Team Robocon 2024, IIT BHU</i>	<i>Report</i>
• Developed autonomous agricultural bot using ROS and PID control for seedling and paddy rice handling • Implemented robot perception using 2D LIDAR, IMU, and computer vision for crop detection and navigation • Integrated SLAM and Tele-operations on Jetson Nano and Raspberry Pi for system control • Designed mechanical systems and optimized bot performance through Gazebo simulation and CAD modeling • Exposure: Python, Gazebo, ROS, Robotics Control, Segmentation, PID, CAD (Fusion 360), Computer Vision	
Path Planning and Execution for a PX4 Iris Drone	Sep 2024 - Oct 2024
<i>Inter IIT Prepathon, IIT(BHU)</i>	<i>Solution</i>
• Simulated PX4 drone using MAVROS and ROS 2 for communication and implemented Gazebo for testing • Developed autonomous navigation system with real-time obstacle avoidance and waypoint optimization • Integrated YOLO-based computer vision system for weed detection and segmentation using drone camera feed • Implemented end-to-end pipeline combining path planning, image processing, and autonomous control • Exposure: PX4 Autopilot, MAVROS, ROS2, Gazebo, YOLO, Path Planning, OpenCV, Python, C, C++	

Advanced Cobots for Logistics in Smart Warehouse

Sep 2024 - Present

E-Yantra Competition 2024-2025, IIT Bombay

[Report](#)

- Developing intelligent dual-cobot system using **ROS2** and **NAV2** for automated warehouse operations
- Implemented real-time package identification and sorting using **computer vision** and **sensor fusion** techniques
- Created adaptive autonomous navigation system with **path optimization** for efficient package delivery
- Integrated advanced **control systems** for seamless coordination between RoboArm and RoverBot cobots
- **Exposure:** ROS2, NAV2, Control System, Robotics, Automation, Path Optimization, Sensor Integration

Advanced Integrated Robotic Basketball System

Oct 2024 - Present

Team Robocon 2025, IIT (BHU) Varanasi

[Report](#)

- Completed base design and **CAD modeling** for two-robot with pneumatic shooting mechanism
- Implementing **computer vision** system for basket detection and trajectory optimization for accurate shooting
- Developing **multi-robot coordination** system with real-time communication for strategic gameplay
- Integrating **control systems** for automated dribbling, passing, and defensive maneuvers between robots
- **Exposure:** ROS, Robotics, Computer Vision, CNN, YOLO, Control Systems, Team Coordination

Text/Video to Advanced Sign Language Conversion

Jan 2025 - Present

EXPLORATORY Project Under Prof. Dr. Shyam Kamal

[Report](#)

- Completed data processing pipeline using **computer vision** for sign language gesture extraction from video inputs
- Implementing **Large Language Models (LLMs)** for text-to-sign language conversion with enhanced accuracy
- Developing extensive avatar system using advanced **3D modeling** to generate realistic sign language animations
- Will integrate interactive web application with secure **REST API** for model deployment and user interaction
- **Exposure:** Machine Learning, API Integration, LLM, Data Processing, Web Application Development

Rotary Inverted Pendulum Control

Jan 2025 - Present

Personal Project

- Designed and assembled a rotary inverted pendulum using **3D-printed components**
- Implemented a **PID controller** for initial stabilization and control of the pendulum
- Exploring advanced control techniques like **Linear Quadratic Regulator (LQR)** and modern control strategies
- Planned future **reinforcement learning (RL)** implementation to improve adaptive control and performance
- **Exposure:** Control Systems, PID, LQR, IMU Filtering, Reinforcement Learning, Robotics, Signal Processing

HONOURS AND ACHIEVEMENTS

- Successfully advanced to Stage 2 and exhibited bot in Stage 3 of **Robocon 2024** representing IIT BHU
- Successfully cleared First Stage and actively progressing in Stage 2 of **E-Yantra Competition**, IIT Bombay
- Selected for Remote Research Internship position at **Changwon National University**, South Korea
- Successfully secured overall 8th Rank among all prestigious IITs in **Inter IIT Tech Meet 13.0**

RELEVANT COURSEWORK

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| • MA-102: Linear Algebra and Derivatives | • MA-202: Probability and Statistics |
| • EO-102: Fundamentals of Electronics and Instrumentation Engineering | • Neural Networks and Deep Learning by Andrew Ng - DeepLearning.AI |
| • CSO-101: Computer Programming | • Convolutional Neural Networks by Andrew Ng - DeepLearning.AI |
| • MA-201: Numerical Techniques | • Introduction to Computer Vision and Image Processing by IBM |
| • Introduction to Self-Driving Cars by University of Toronto | |

MEMBERSHIPS

- Project Lead at **Inter IIT TechMeet 13.0**, developing advanced UAV control architectures and failure mitigation algorithms in collaboration with IdeaForge for mission-critical drone operations
- Core Team Member **Robocon 2025**, spearheading robot design and coordinating cross-functional development
- Core member , **Robotics Club**, IIT(BHU) Varanasi, mentoring juniors through workshops
- Member of **Robocon 2024**, representing IIT(BHU) and managing cross-functional team operations